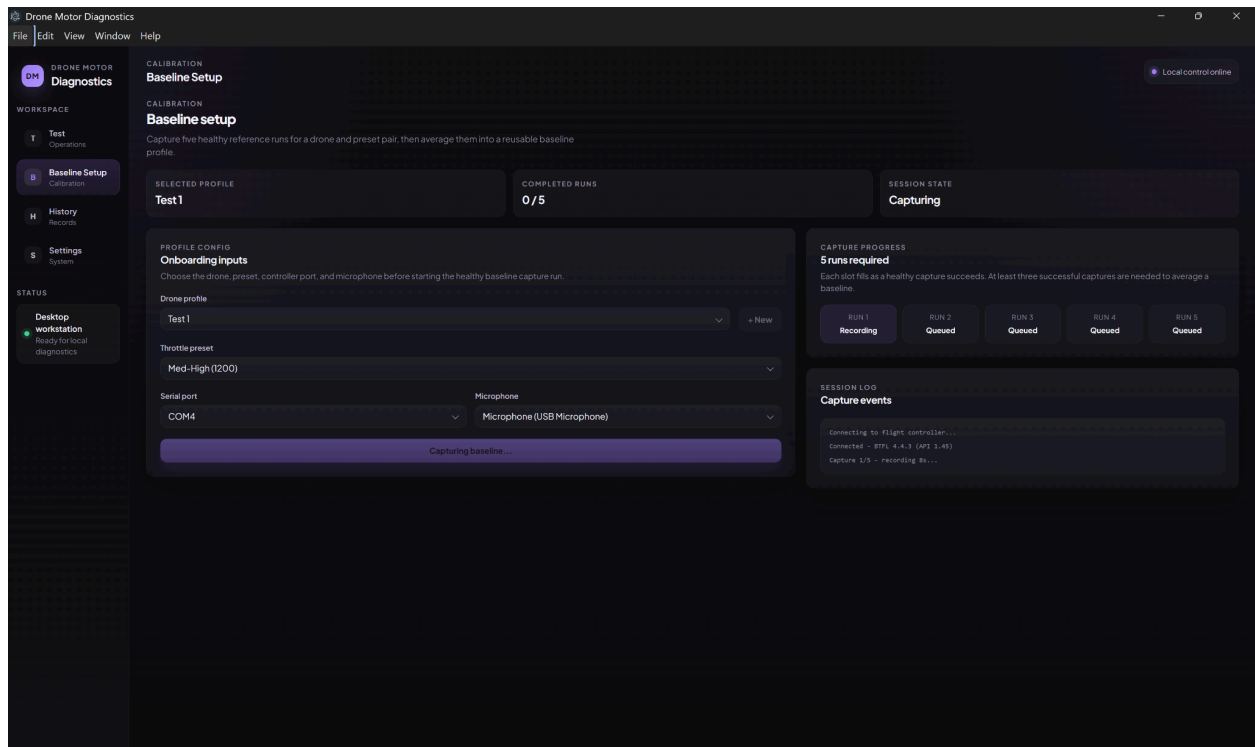
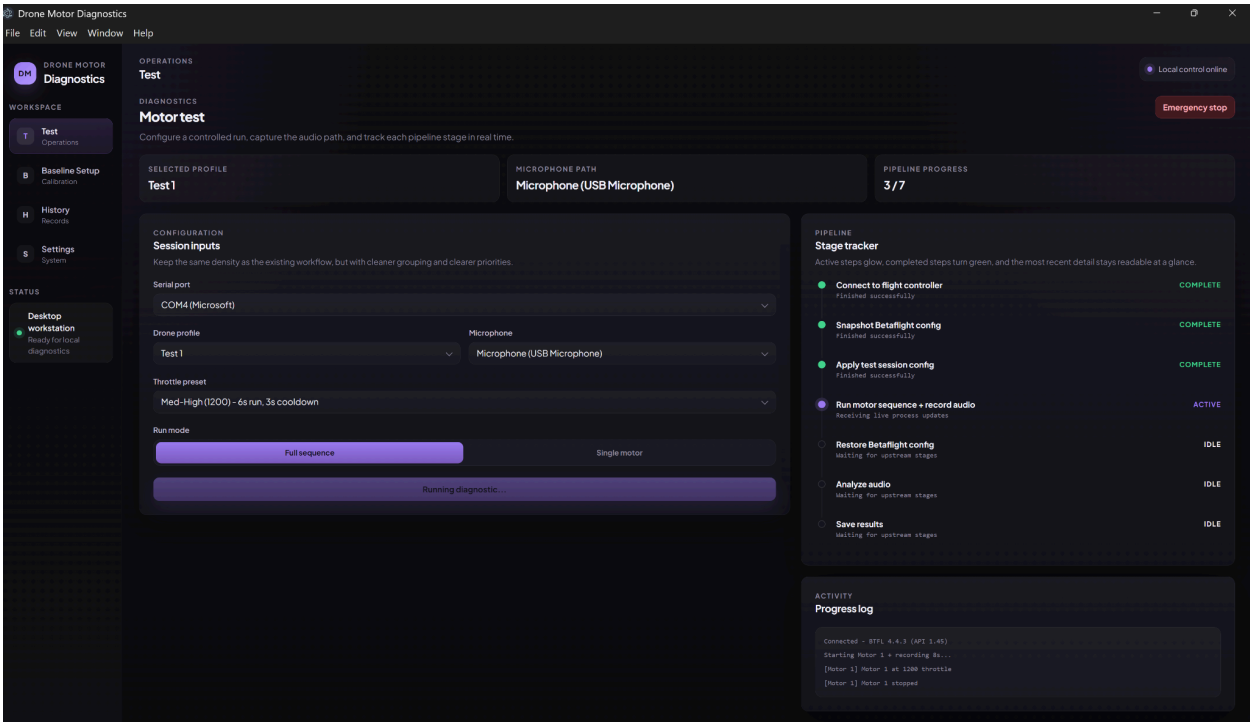
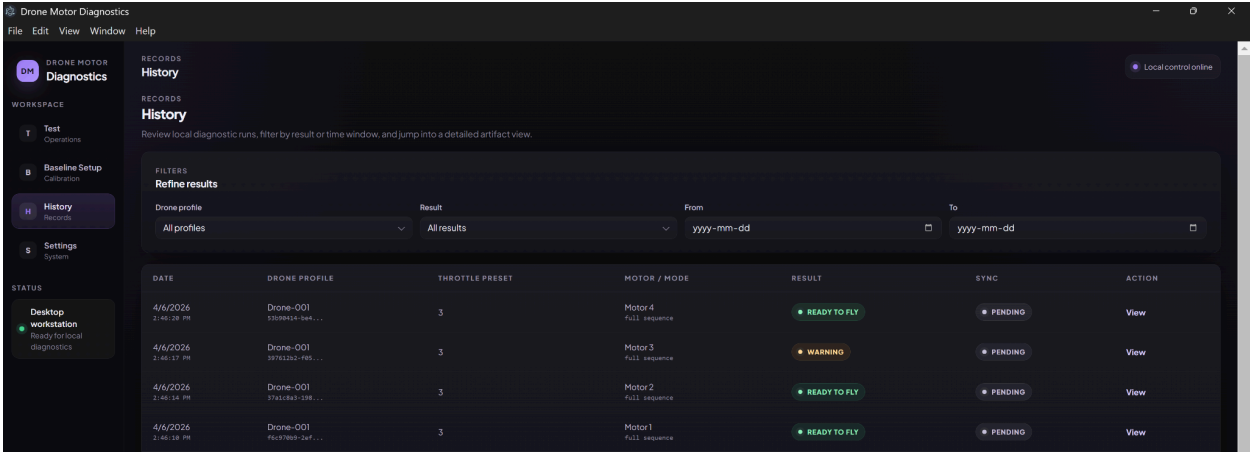


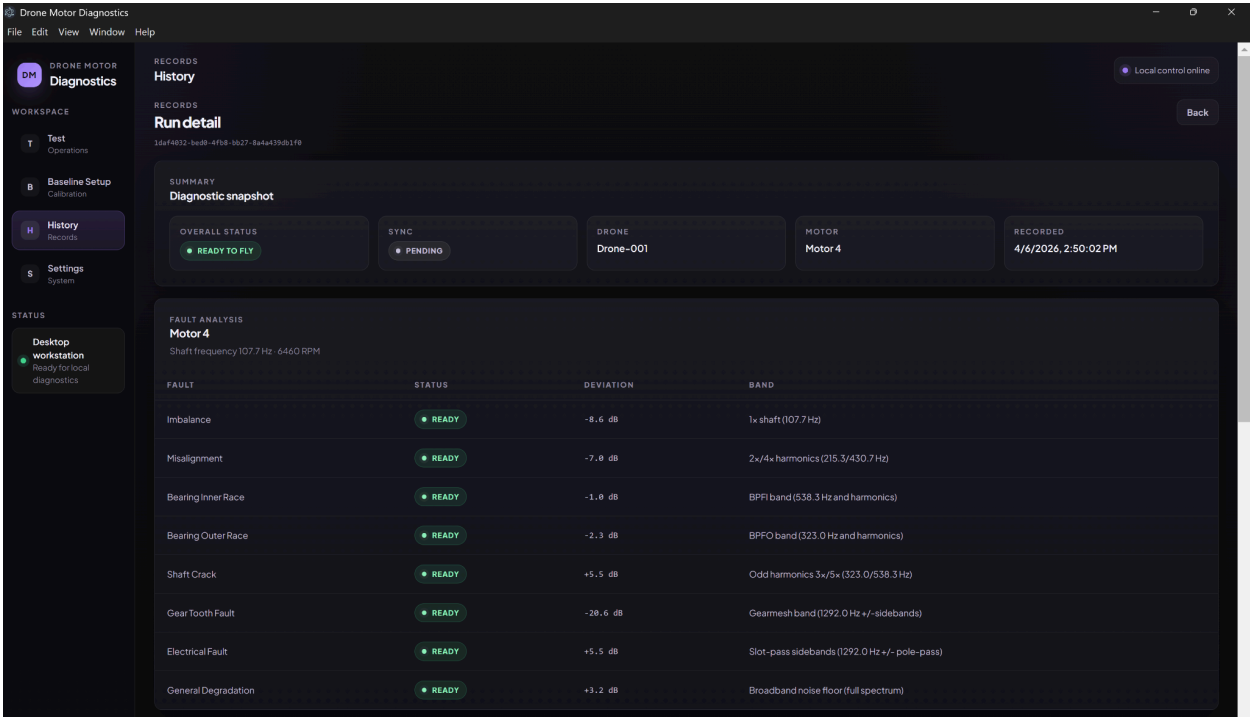
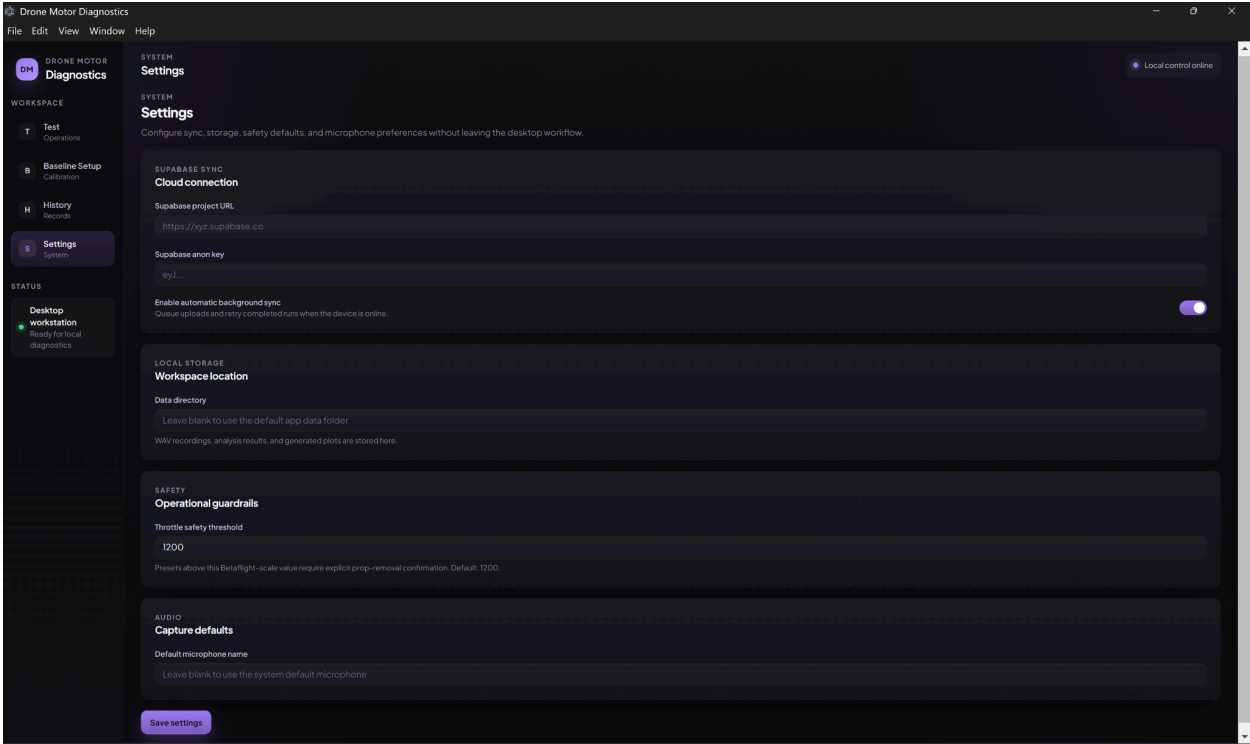
One requirement to add:
How in depth is the data showing failure of the drone
Various points of failure (the more the better)

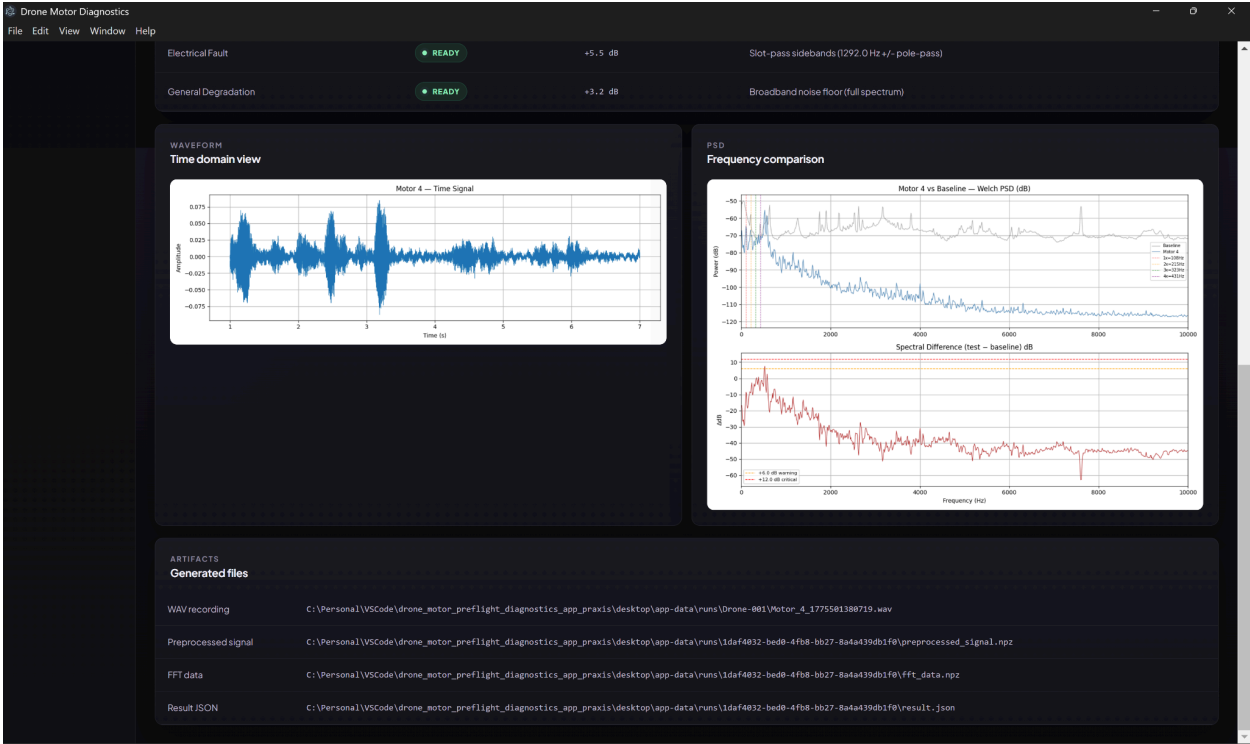
Need test protocol
Measurement Matrix
Pugh chart

Studio feedback
Explain the concept, the idea (input -> output), sketches, visual representation
Showcase: all three concepts, full FDCR
25% framing, 25% diverging, 25% converging 25% verification, test, final result









SUMMARY

Diagnostic snapshot

OVERALL STATUS

WARNING

SYNC

PENDING

DRONE

Drone-001

MOTOR

Motor 3

RECORDED

4/6/2026, 2:46:17 PM

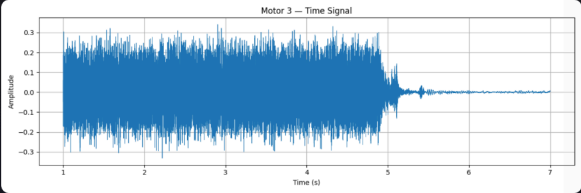
FAULT ANALYSIS

Motor 3

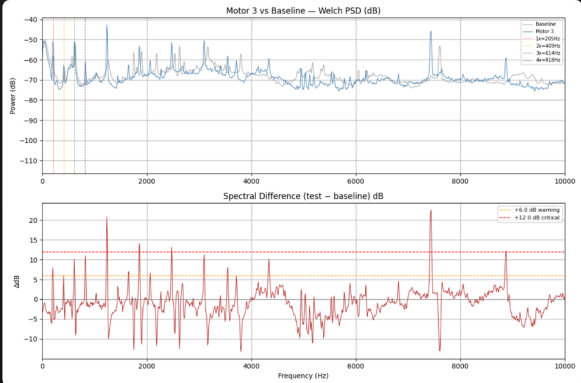
Shaft frequency 204.6 Hz · 12274 RPM

FAULT	STATUS	DEVIATION	BAND
Imbalance	WARNING	+8.0 dB	1× shaft (204.6 Hz)
Misalignment	WARNING	+10.0 dB	2×/4× harmonics (409.1/818.3 Hz)
Bearing Inner Race	READY	-0.9 dB	BPFI band (1022.8 Hz and harmonics)
Bearing Outer Race	READY	-0.8 dB	BPFO band (613.7 Hz and harmonics)
Shaft Crack	READY	+1.5 dB	Odd harmonics 3×/5× (613.7/1022.8 Hz)
Gear Tooth Fault	READY	-0.9 dB	Gearmesh band (2454.8 Hz +/- sidebands)
Electrical Fault	READY	+0.9 dB	Slot-pass sidebands (2454.8 Hz +/- pole-pass)
General Degradation	READY	+2.4 dB	Broadband noise floor (full spectrum)

WAVEFORM
Time domain view



PSD
Frequency comparison



ARTIFACTS
Generated files

WAV recording	C:\Personal\VSCode\drone_motor_preflight_diagnostics_app_praxis\desktop\app-data\runs\Drone-001\Motor_3_1775501146238.wav
Preprocessed signal	C:\Personal\VSCode\drone_motor_preflight_diagnostics_app_praxis\desktop\app-data\runs\397612b2-f055-4857-b168-f7d17550e3ff\preprocessed_signal.npz
FFT data	C:\Personal\VSCode\drone_motor_preflight_diagnostics_app_praxis\desktop\app-data\runs\397612b2-f055-4857-b168-f7d17550e3ff\fft_data.npz
Result JSON	C:\Personal\VSCode\drone_motor_preflight_diagnostics_app_praxis\desktop\app-data\runs\397612b2-f055-4857-b168-f7d17550e3ff\result.json

Fault 1: Mass Imbalance

Spectral signature: Dominant peak at $1\times$ shaft rotation frequency (fundamental), with harmonics. The peak at the fundamental grows significantly relative to baseline.

Source: *Introduction and Background*, p. 22; *Vibration Signals from Rotating Machines*, Ch. 2

Quote: "sinusoidal components in response signals result from sinusoidal forcing functions at the same fundamental frequency...the responses will usually be distorted from sinusoidal and thus contain some level of harmonics of the fundamental frequency, even if the forcing function is relatively pure, such as a **simple unbalance**."

What to look for on your Welch PSD: A large dB increase at the motor's fundamental rotation frequency ($1\times\text{RPM}/60\text{ Hz}$) compared to baseline.

Fault 2: Misalignment

Spectral signature: Strong increase at the 2nd and 4th harmonics of shaft speed. Misalignment characteristically excites higher harmonics rather than just the fundamental.

Source: *Fault Detection*, pp. 151–152 (Figure 4.6 and surrounding text); *Diagnostic Techniques*, Figure 5.28

Quote: "it was confirmed that it coincided with the speed of the compressor shaft and that the **second and fourth harmonics were quite high**...the increases in vibration were due to increasing **misalignment**."

What to look for: dB increases specifically at $2\times$ and $4\times$ the fundamental shaft frequency, disproportionate to the $1\times$ peak.

Fault 3: Rolling Element Bearing Fault (Inner Race)

Spectral signature: Impulses at the Ball Pass Frequency Inner race (BPFI), with sidebands spaced at shaft speed around the BPFI harmonics. These appear at high frequencies (can be $1000\times$ shaft speed), and are often only visible after envelope analysis of a high-frequency band.

Source: *Diagnostic Techniques*, pp. 195–196, 211–212 (Section 5.5, Figure 5.51)

Quote: "the benchmark method for bearing diagnostics is **envelope analysis**, where a signal is bandpass filtered in a high-frequency band in which the fault impulses are amplified by

structural resonances...whose spectrum contains the diagnostic information in terms of both repetition frequency (**ballpass frequency**) as well as modulation by the appropriate frequency."

And more specifically: "a typical envelope spectrum for an inner race fault, with a **series of harmonics of BPFI** (1398 Hz), together with low harmonics of, and **sidebands spaced at, shaft speed** (196 Hz)."

What to look for: New peaks appearing at BPFI and its harmonics, with sideband clusters around them spaced at the shaft rotation frequency.

Fault 4: Rolling Element Bearing Fault (Outer Race)

Spectral signature: Impulses at Ball Pass Frequency Outer race (BPFO). Unlike inner race faults, outer race faults are typically *unmodulated* (no sidebands at shaft speed) when the load is unidirectional, because the fault always passes through the load zone at the same position.

Source: *Diagnostic Techniques*, p. 205 (Section 5.5.1)

Quote: "For **unidirectional load, an outer race fault would not be modulated**, but modulation at shaft speed can occur because of significant unbalance or misalignment forces."

What to look for: Clean harmonics of BPFO without the shaft-speed sideband clusters that characterise inner race faults.

Fault 5: Shaft Crack (Breathing Crack)

Spectral signature: Appearance of odd harmonics of shaft speed (1×, 3×, 5× etc.) that were not present in the baseline. A permanently open crack generates even harmonics; a breathing crack (opening and closing each revolution) generates odd harmonics.

Source: *Introduction and Background*, p. 22; *Vibration Signals*, Ch. 2, Section 2.2.1.2

Quote: "In the case of a **crack in a shaft**...if the crack is 'breathing', that is opening and closing every revolution of the shaft, the character of the forcing function changes, **giving rise to responses at the odd harmonics of the shaft speed**, in contrast to the even harmonics primarily generated when the crack is permanently open."

What to look for: New spectral peaks appearing at odd multiples of shaft frequency (3×, 5×) that were absent in the baseline.

Fault 6: Gear Tooth Fault (Localized — Crack or Spall)

Spectral signature: Sidebands spread widely around the toothmesh frequency (shaft speed $\times$ number of teeth) and its harmonics. A localized fault produces a large number of sidebands spaced at shaft speed, spread across a wide frequency range.

Source: *Fault Detection*, p. 153 (Figure 4.9); *Diagnostic Techniques*, Section 5.4

Quote: "the spectral changes are primarily at **widely distributed harmonics** of a frequency (which corresponds to the speed of the input pinion), most of which can be interpreted as **sidebands around the first two harmonics of the gearmesh frequency**. Such a wide distribution of sidebands is **typical of a localized fault on a gear**."

What to look for: Proliferation of sideband peaks spreading symmetrically around the toothmesh harmonic peaks, spaced at shaft rotation frequency.

Fault 7: Electric Motor Rotor/Stator Fault (Eccentricity / Winding)

Spectral signature: Sidebands around the slot-pass frequency (number of rotor slots $\times$ shaft speed) and around twice the mains/electrical supply frequency, spaced at slip frequency $\times$ number of pole pairs.

Source: *Fault Detection*, p. 148; *Diagnostic Techniques*, p. 161

Quote: "higher harmonics such as gearmesh frequencies, blade-pass frequencies, and **slot-pass frequencies in electric motors...the sideband spacing is at slip frequency times the number of poles**."

What to look for: Sideband clusters appearing around the electrical commutation frequency (visible in your Motor 1 spectrum as the ~2000 Hz peak), with spacing related to slip frequency.

Fault 8: General Mechanical Degradation (Raised Noise Floor)

Spectral signature: Broad increase in the spectral noise floor across many frequencies simultaneously, rather than specific new peaks. This is a non-specific indicator that something has changed but can't yet be localized.

Source: *Fault Detection*, p. 140 (Section 4.2.1)

Quote: "it is advisable in general to **detect changes in vibration spectra**, rather than overall RMS levels...Cases abound where significant changes in individual frequency components **only affect the overall vibration levels at the very last stages**, if ever."

What to look for on your Welch PSD: The noise floor between existing peaks rises uniformly across a band — the difference plot compared to baseline will show a broad "shelf" of elevated dB rather than isolated spikes.

Summary Table

Fault	Spectral Indicator	Key Frequency
Imbalance	Large peak at 1× shaft freq	1× RPM/60
Misalignment	Elevated 2× and 4× harmonics	2×, 4× RPM/60
Bearing inner race	BPFI harmonics + shaft-speed sidebands	Depends on geometry
Bearing outer race	Clean BPFO harmonics, no sidebands	Depends on geometry
Shaft crack (breathing)	New odd harmonics (3×, 5×)	3×, 5× RPM/60
Gear tooth fault	Wide sideband spread around toothmesh	$N_{\text{teeth}} \times \text{RPM}/60$
Motor electrical fault	Sidebands around slot-pass / 2× mains	Slots × RPM/60
General degradation	Raised broadband noise floor	Broadband

This gives you a solid framework for the detection logic. Once you know your motors' RPM we can calculate the exact expected frequencies for each fault type and wire them into the comparison code.